

June 22, 2026

Dr. Emilie Chouzenoux
OPIS Team
Inria Saclay-Île-de-France
Paris, France

Dear Dr. Emilie Chouzenoux,

I am writing to express my strong interest in the PhD position focusing on robust few-shot learning for foundation models in CT imaging (ANR FM2AI project). As an AI & Data Science Engineer specializing in medical deep learning and rigorous mathematical modeling, the opportunity to advance tumor segmentation and predictive diagnostics across 50,000 CT scans aligns perfectly with my research trajectory.

During my Master's thesis on the "Cardiovascular Medical Digital Twin," I engineered complex architectures capable of reasoning over sparse, multi-modal clinical data. Specifically, I developed Medical_MambaTS, a linear-time state-space model capable of handling long-context sequences without the quadratic bottleneck of standard transformers. More importantly, I engineered PI-DeepONet, a Physics-Informed Operator Network that explicitly maps the mathematical constraints of hemodynamics (0D Windkessel) into the neural architecture via PyTorch Autograd.

My experience embedding strict structural bounds directly into the neural loss function is highly applicable to your goal of ensuring the robustness and generalization capabilities of few-shot models. In clinical settings where data is inherently sparse and annotation efforts are costly, ensuring that few-shot adaptation does not mathematically degrade or hallucinate is paramount. I am eager to apply my background in continuous-time modeling and rigorous convergence analysis to the transductive few-shot paradigms your team at OPIS is pioneering.

I possess extensive experience programming in Python/PyTorch, and I am highly motivated by the interdisciplinary and clinically focused environment of the FM2AI consortium.

Please find attached my CV, academic references, and summaries of my engineering theses. I would welcome the opportunity to discuss my qualifications with you in an interview.

Sincerely,
Ismail Aissaoui
<https://ismail-aissaoui.vercel.app>